When computers call the shots — exploring the impact of AI in high-stakes decision-making

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Abstract

While applications abound for the modern surge of artificial-intelligence (AI)-based technologies, we owe special care to the ones where *human lives are at stake*. Perilously, AI advances faster than the regulations that protect us from it. As such, in this review, **we explore the impact of AI on high-stakes decision-making.** Though reviews have attempted adjacent analyses before, none have effectively targeted safety-critical industries or considered the relation between AI developers, users, and regulators. To address these gaps, we conduct a systematic literature review. We cast a wide net in our search, then filter with a series of criteria that qualify a system's sensitivity, decision-making framework, and risk. After distilling our initial pool of 1,135 papers into a final subset of 72 reports, we identify critical features of AI systems and industry implementation practices that demonstrate: 1) the safety risks of using AI in high-stakes decision-making, 2) how we can mitigate these risks, and 3) how we can assign responsibility for AI-based decisions. A major finding of our review is that, contrary to popular broad-calls for increasing model interpretability and explainability, we find these features only useful in specific safety-critical scenarios– when the user has both the *expertise and time* to review a model output. Such cases do not require regulation to authorize user trust, and we expect overregulating these cases to actually be counterproductive. Alternatively, when expertise or time is limited, validation (*not* explanation) techniques and robust regulation *must* authorize user trust. This review bridges the gap between developers, users, and regulators on safety-critical AI applications and makes specific recommendations for policy and future work based on its findings.

Keywords: AI-Informed Decision-Making, Sensitive Industries, Systematic Literature Review, AI Regulation

1. Introduction

There is no more critical time to evaluate the safety-risks artificial intelligence (AI) poses to our most sensitive industries than when governments are actively devising strategies for how regulate AI [1]. To properly mitigate these risks, we need to understand how these systems are applied, who they are used by, and what kind of autonomy they have. Then, by evaluating the gaps between developers, users, and regulators, we can piece together a more complete picture of the high-stakes AI environment, and appropriately recommend safeguards. Other authors have attempted to characterize this landscape in the past, but have missed key features. For example, Wang and Chung [2] perform a systematic literature review on AI in safety-critical industries, but fail to target specific industries, do not distinguish between validation and explanation, and neglect the impact of regulation. Additionally, Bach et al. [3] perform a systematic literature review on human-AI interaction in safety-critical industries, but again, make broad calls for explainability without considering the application, fail to discuss validation methods, do not target specific industries, and miss the mark on establishing trust frameworks.

1.1. Novelty

To fill this gap, we conduct a systematic literature review on AI-based decision-making in sensitive industries. As seen in [2] and [3], the "black-box" nature of AI tends to take the brunt of the blame for model mistakes and induced accidents. While calls for transparency are all in good faith, we argue that a broad "explainability-based" solution may not suit all applications. This review establishes the differences between those applications through a generalizable framework by considering a wide variety of industries.

While the safety outcomes we consider all include some threat to human health or safety, the diversity of these industries means that use of AI may risk the health of one to many individuals with low to high probability. This variety supports our conclusions as generalizable. Aside from the size of the affected population and the likelihood of an incident, the applications include variety in *who* interacts directly with the AI system in question and the time they have to decide (or intervene). Our study brings novelty to the field by extracting key qualities of sensitive industries, understanding how AI affects outcomes based on those qualities, and identifying how those interactions impact safety, trust, and liability. Further, we make recommendations for regulators, users, and developers based on these findings.

2. Method

This review prioritizes reproducibility and therefore collects documents in a systematic and automated way. To do so, we start by defining the research questions this review seeks to answer. These include:

1. *What is the impact of AI on high-stakes decision-making?*
2. *What are the safety risks of using AI to make decisions in sensitive industries?*
3. *What happens when accidents occur as a result of an AI-based decision?*

These questions distinguish the unique features and outcomes that set AI-based decision-making mechanisms apart from traditional methods. Likewise, answers would provide insight for both governmental agencies and engineers to mitigate evident shortcomings of this high-impact technology.

2.1. Query Term Compilation

After determining our research questions, we define a set of key terms using the Population, Intervention, Comparison, Outcome (PICO) methodology [4], and use the Cartesian product of these terms (one from each category) to determine relevant search queries. We use APIs for each of two databases, Scopus and Web of Science, to make these queries via a Python workflow, storing all resultant papers. Then, we remove duplicates based on digital object identifier (DOI), and merge the results of both database searches into a combined bibliography. Figure 1 shows visual representation of this workflow.

Critical to actually making these queries is defining the lists of keywords that comprise them. We start by defining our *population* terms.

We can organize the population terms we decided to include in this search in five major groups:

- autonomous driving,

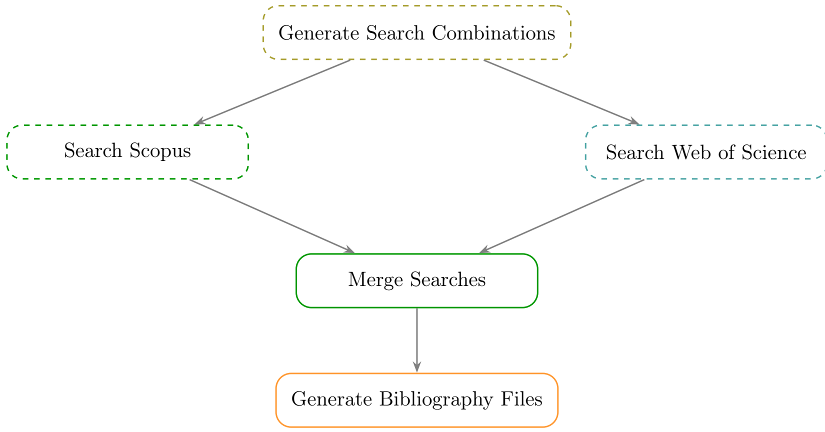


Figure 1: Directed acyclic graph showing automatic search workflow.

- aviation,
- energy and industrial processes,
- medical technology and pharmaceuticals, and
- nuclear and radiation.

These groups were selected for their highly sensitive nature, their propensity to use AI, and their regulatory authorities. This includes oversight by the National Highway Traffic Safety Administration (NHTSA), the Federal Aviation Administration (FAA), the Occupational Health and Safety Administration (OSHA), the Food and Drug Administration (FDA), and the U.S. Nuclear Regulatory Commission (NRC), in the United States. Though we mention these agencies explicitly, our review considers regulation across international borders as well, including oversight from the EU, United Kingdom, and Japan.

Though incomplete, we prioritize the industries listed above as a representative set in our analysis. In determining our search terms, we tried to avoid cases that consider risks *not imminently* related to human health and safety (financial risks, environmental risks); we will discuss this further when we consider our exclusion criteria. The population search terms we include in our querying are shown in Table 1.

After population, we must consider our *intervention* terms. We determined this group by compiling a list of AI-related terms that would likely appear in any AI-based decision-making model. This includes words and phrases like "machine learning," "neural networks," and "language models." In our full query, which we describe later in this section, we use the search operator `W/5` and `NEAR/5` for Scopus and Web of Science, respectively, to pair each intervention term in Table 1 with the wildcard term "decision*". In lieu of a mere `AND` operator, these operators require the word "decision" and its derivatives to be within five words of the AI-intervention term. The objective of this pairing is to return articles that use AI to either make or inform some kind of decision. The intervention search terms we include in our querying are shown in Table 1.

Finally, we consider our *outcome* terms. Our research question focuses on the impact of AI-based decision-making on safety outcomes in sensitive industries. Thus, our outcome terms focus on safety, specifically, on risks to human health. We include general terms like "incident," "accident," and "near miss," as well as more industry-specific terms, like "misdiagnos*" and "operating experience." To capture any failures that led to official infractions, we included several regulation-specific terms, too, like "FDA recall*" and "OSHA citation." We do not consider *comparison* terms in the search term compilation, but we do exclude certain kinds of comparisons during screening (i.e., AI vs AI instead of traditional method vs AI). A complete list of outcome terms used in our querying are shown in Table 1.

2.2. Search Process

We searched two databases, Scopus and Web of Science, for all articles containing our three primary search terms in their title, abstract, or keywords. We imposed two additional requirements, that the papers be published after and not including 2014 and that the papers either be journal articles or conference papers from the top five AI conferences as listed by Google Scholar Metrics [5]. While we assume journal articles will sufficiently cover most domains, some relevant computer science appears in conference papers only. This conference paper subset includes *Neural Information Processing Systems* (NEURIPS), the *International Conference on Learning Representations* (ICLR), the *International Conference on Machine Learning* (ICML), the *AAAI Conference of Artificial Intelligence* (AAAI), and the *International Joint Conference on Artificial Intelligence* (IJCAI). We created this subset using the combination of both a paper type of `conference proceedings` and that conference proceeding containing one of the five acronyms listed above. In the case that this search catches an additional conference, that paper will be excluded during the abstract screening phase.

We combined our search terms into the following query search:

```
TITLE-ABS-KEY ( {population term} ) AND TITLE-ABS-KEY (
{intervention AI term} within five words of {decision*})
AND TITLE-ABS-KEY ( {outcome term} ) AND PUBLICATION YEAR
AFTER 2014 AND (DOCUMENT TYPE(article) OR (DOCUMENT TYPE(conference
proceeding) AND {conference subset}))
```

Note: the query format above matches Scopus syntax most closely, but is intended for demonstrative purposes only. We show our exact query structure in the code repository. The final set of articles considered for this analysis was pulled from both Scopus and Web of Science on **December 15, 2025**.

As shown in Figure 1, after searching the full set of queries in both Scopus and Web of Science, merging the search results into a single database and dropping duplicates based on DOI, we generated a singular bibliographic file in the `.bib` structure. We then loaded the bibliographic information into Zotero [6]. We removed two records Zotero marked retracted in the pre-screening phase along with one manually identified duplicate record. This yielded a total of 630 records for screening. Figure 2 shows the number of articles retained at each phase of the filtration process.

2.3. Screening Process

Following the PRISMA methodology [7], we perform the screening process in two phases– title and abstract screening, and full-text review. During the abstract screening, we followed the inclusion/exclusion criteria shown in Table 2, keeping questionable articles conservatively based on what we read in the titles and abstracts *only*. The purpose of this phase is to quickly remove all articles that absolutely do not meet our criteria, while keeping any that may. During the abstract screening, this mainly involved removing articles with an emphasis on outcomes that were not threats to human health or safety.

We excluded many articles during this phase because they fell under the category of "predictive maintenance," which we define as using AI to reduce the frequency of scheduled maintenance. By design, these studies focus on reducing costs and, at best, have no impact on safety. Systems can conservatively avoid the worst case safety impacts by following their original

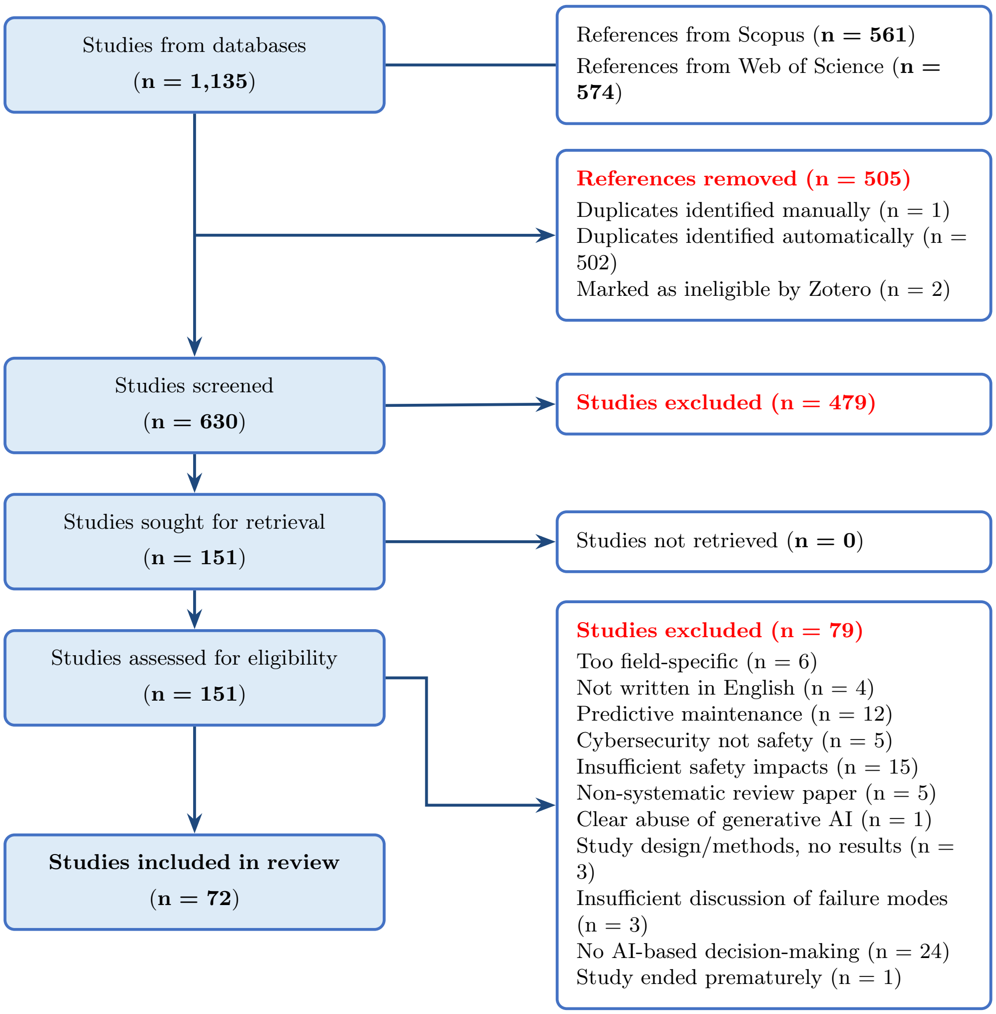


Figure 2: Paper filtration flow diagram based on PRISMA [7].

maintenance schedule. Note that when AI used to predict equipment failures, without reducing maintenance frequency (such as ML-based alarms), we do not consider this predictive maintenance and include these studies in our review.

While these threats are dangerous, we also excluded papers that focused on cybersecurity and, in turn, did not imminently threaten human health. To maintain our scope, we focused on physical and immediate threats. Future work, however, should consider security risks of AI-based decision-making.

We also created exclusion criteria to define "AI." Because we sought to consider regulatory approaches to AI in this analysis, we drew the line at pure linear regression based models and IF/ELSE-type logic models, since these algorithms are already widely accepted. Finally, we excluded any reviews that were non-systematic, any articles that were retracted, any ethics-based papers without a clear application to the AI technology or use of AI technology in their target domain, and any papers not written in English. To assist with our screening process, we used Covidence, a website designed for systematic reviews [8]. We did not track case-by-case reasons for exclusion during the abstract screening phase. As shown in Figure 2, we excluded 479 articles in this phase, yielding 151 articles for the full-text review process.

2.4. Full Text Screening

After the title and abstract screening, we proceeded to the full text screening stage. This phase, as its name suggests, involves reading the full text of each paper and applying the same inclusion/exclusion criteria shown in Table 2. During the full-text screening phase, we track the exclusion reason for each article. The number of articles and their specific exclusion reason are shown in Figure 2.

We did include some additional exclusion reasons during this phase to manage scope. One example was: "too field-specific." This was only a handful of papers, but consisted of studies that overly focused on the AI algorithm or the application, and insufficiently described their intersection. Similarly, we did not include methods-only papers. We excluded one paper due to clear abuse of generative AI in its creation with an apparent lack of editing or attribution, and removed this paper from all materials for its credibility concerns. We could access the abstracts of some articles in English, but not the full text, so two studies were excluded this way as well.

Overall, we attributed the majority of exclusions to a lack of decision-making frameworks in the actual application of the AI. While we allowed for a variety of decision-making types (autonomous, augmented human-decision-making, human-in-the-loop, and even static benchmark setting), if the AI outputs were not applied for *any* type of decision-making, we excluded it. The goal of this was, again, to highlight the most safety critical scenarios that involve AI. If no AI-based decision-making takes place, it becomes challenging to attribute any of the outcome towards the AI. During the full-text review, we excluded 79 articles. The remaining 72 articles, as shown in Figure 2 comprise the final set included in this review.

2.5. Data Extraction

Finally, we extracted the data from the final subset of 72 articles, which included tabulating data for a variety of fields: industry type, industry subcategory, AI type, comparison type, inclusion of explainability and type, consideration of regulation and regulatory authority, decision-making type, data type, whether the system was deployed, output feature considered, safety outcome, and country of origin. Note: while the term "model explainability" has a variety of definitions, here we assume the most broad– anything that constitutes unpacking the logic of an AI model (interpretability, feature importance, etc.) counts. We do not consider, however, the sheer mention of a "black-box" as consideration of explainability.

2.6. Quality Assessment

During the data extraction, we also conducted an abbreviated quality assessment. Due to the inhomogeneous nature of the types of papers included in this analysis, a single, premade quality assessment template would not allow sufficient flexibility for our set of articles. Accordingly, instead of confirming if each article meets a certain set of criteria, we devised a set of "flags" to mark areas of concern for each article based on a pre-existing framework, APPRAISE-AI [9]. Because this review considers a large portion of medical papers, we evaluated each study for its consideration of bias, a known concern for AI applications in that field. To keep our assessment fair, we applied the same judgment for all other papers. Considering all the papers concern human health, the way the AI views the differences between humans can substantially impact its judgment. Other than assessing for bias, we only flagged other categories if we had a concern while reviewing the articles. Commonly, this included flagging for limited dataset size. Most articles scored above 80 out of 100, so there were no major causes for concern regarding quality of the articles retrieved. Figure 3 shows the distribution of articles in the final subset.

3. Quantitative Results

3.1. Querying Results

Figures 4a and 4b show quantitative querying results (the number of results per query combination) from Scopus and Web of Science when indi-

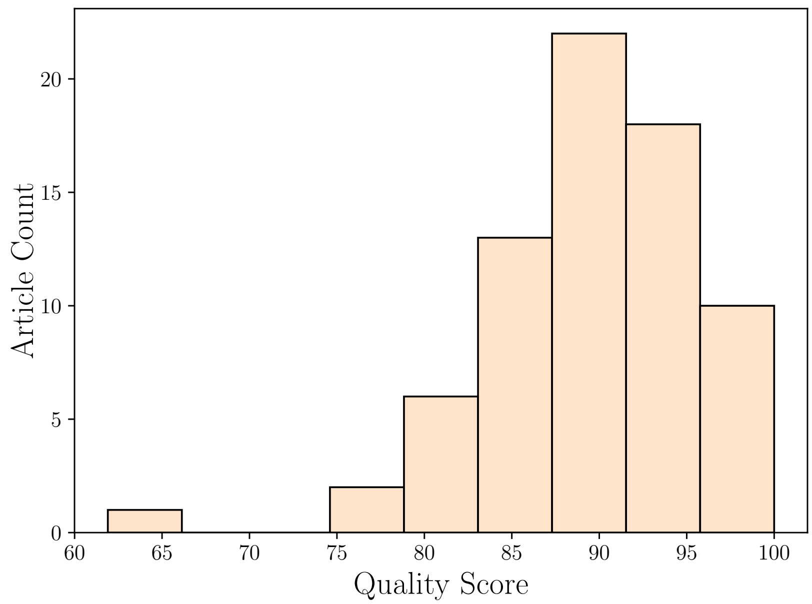


Figure 3: Quality assessment score distribution for articles in the final subset.

vidual search terms were included in all queries.

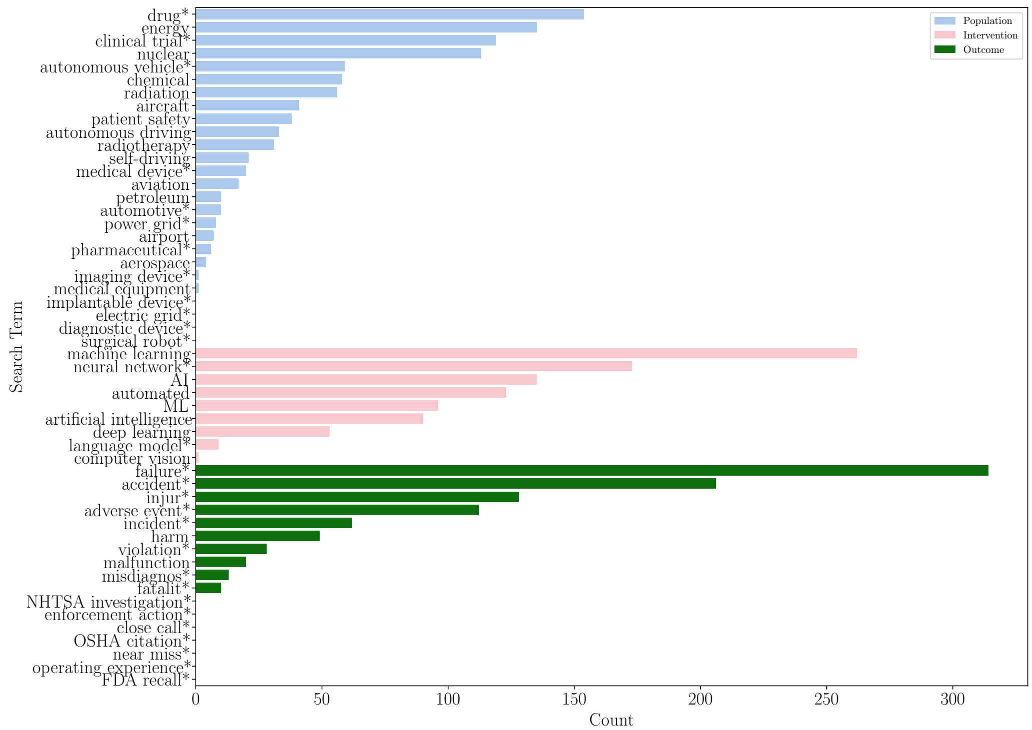
Table 1: Search terms used to produce Cartesian-product-based queries

Population Terms	**Intervention Terms**	**Outcome Terms**
aerospace	"machine learning"	accident*
aviation	ML	"near miss*"
aircraft	"neural network*"	injur*
airport	"language model*"	failure*
"autonomous driving"	computer vision	"adverse event*"
"autonomous vehicle*"	automated	malfunction
self-driving	"deep learning"	fatalit*
automotive*	"artificial intelligence"	misdiagnos*
energy	AI	harm
"power grid*"		"operating experience*"
"electric grid*"		"OSHA citation*"
chemical		"FDA recall*"
petroleum		"NHTSA investigation*"
nuclear		incident*
radiation		"close call*"
"medical device*"		violation*
"medical equipment"		"enforcement action*"
"diagnostic device*"		
"imaging device*"		
"surgical robot*"		
"implantable device*"		
drug*		
"patient safety"		
pharmaceutical*		
radiotherapy		
"clinical trial*"		

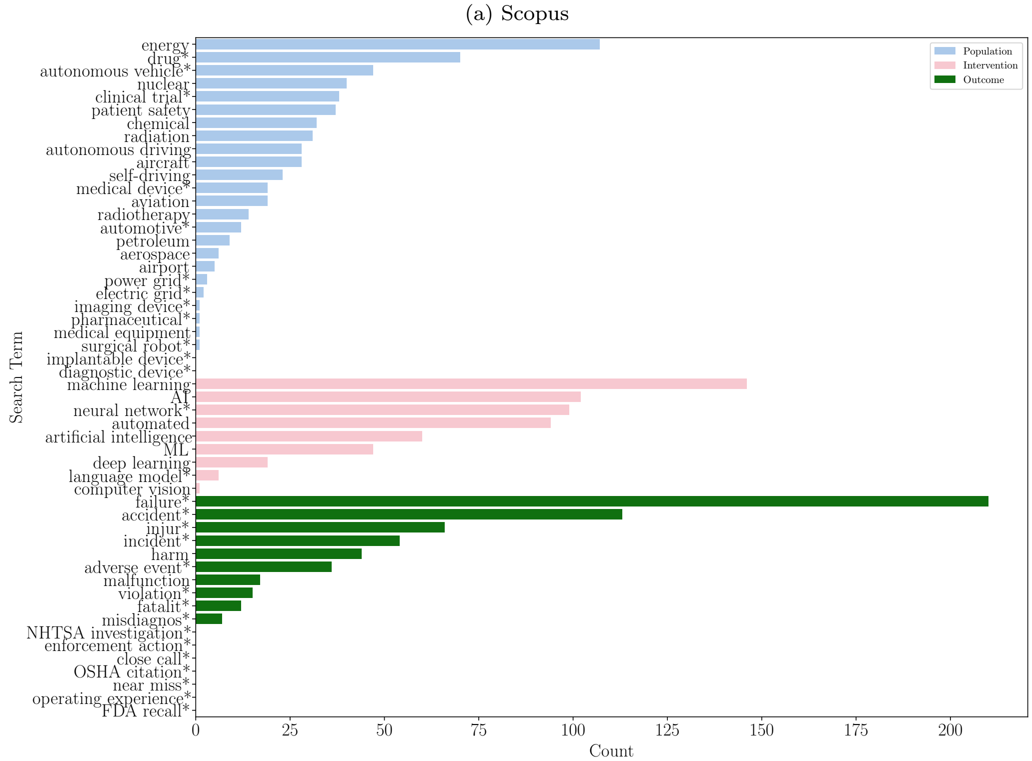
Note: Asterisks (*) indicate truncation wildcards.

	Inclusion	Exclusion
Population	Sensitive technology (e.g., nuclear, aviation) Medical studies, emphasizing equipment, devices, and drugs	Finance Human resources (even if health related) Cybersecurity
Intervention	AI-informed decision-making (with any degree of autonomy)	Simple flow chart or IF/ELSE logic Linear regression models
Comparison	Human evaluation Deterministic method	ML-ML comparison (with exceptions)
Outcome	Incidents, accidents, and failures Threat to human health Regulatory compliance violations	Non-threat to human health risks Financial risks or economic losses Cyber or other security risks
Other	Peer-reviewed journal articles Select conference papers	Non-systematic review papers Retracted articles Ethics without application Not written in English Published before 2015

Table 2: Inclusion and exclusion criteria used for abstract screening phase.



(a) Scopus



(b) Web of Science

Figure 4: Sum of articles returned by each database when a given search term was included in all queries.

3.2. Data Extraction Results

The database used to tabulate our results can be accessed here. Additionally, we provide a database in Supplemental Materials.

After extracting the information mentioned in Section 2, we created figures to portray key trends from the set of articles reviewed. This includes country of author institution, main AI type considered in the article, decision-making type, whether regulation or explainability was considered, and types of explainability methods considered.

Figure 5 shows the number of articles that employed each type of AI in the final subset of articles. Note that many articles employed more than one type of AI, so Figure 5 shows more counts than total articles. Similarly, some articles focused on ethical or regulatory frameworks and therefore considered AI more generally. We denote the AI type in these articles as "Non-Specific."

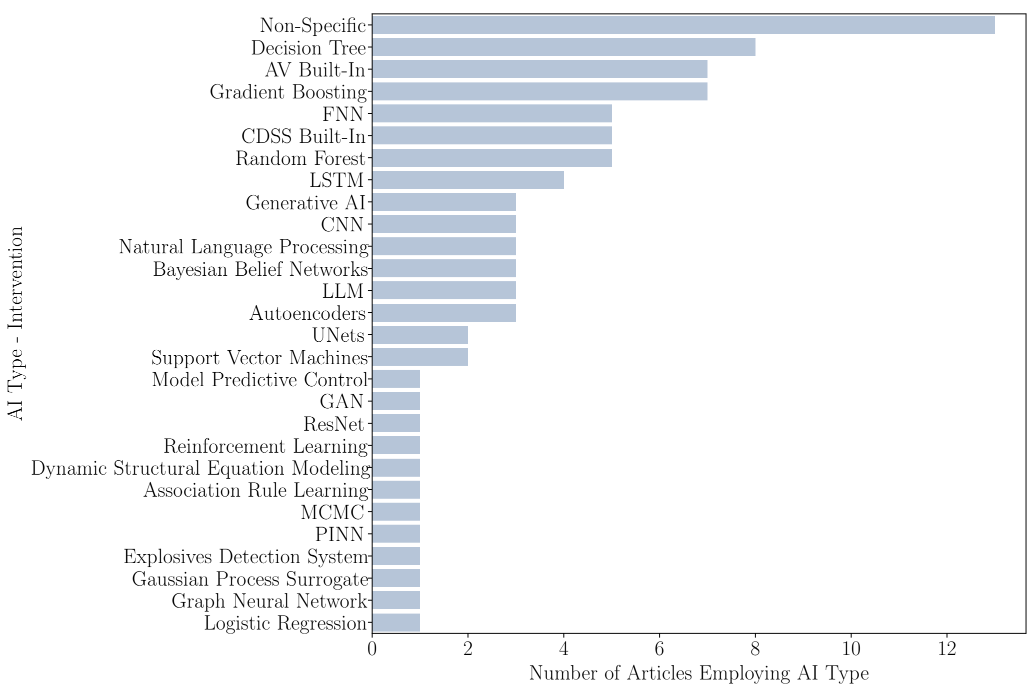


Figure 5: Number of articles in final subset that considered each type of AI method. Non-specific means the article did not provide details about the model type.

Figure 6 shows the decision-making type evaluated by the articles in the final subset. "Not System Integrated" refers to articles that use AI to set some kind of threshold or benchmark value for decision-making, but do not actually integrate a dynamic version of the AI itself into the system under consideration. Additionally, we distinguish "augmented human decision-making" and "human-in-the-loop" as systems that both require human action to make decisions, but human-in-the-loop leans more heavily on the AI system whereas augmented simply uses the AI results as an additional piece of information to make a decision with. Autonomous systems do not necessarily require human oversight to take action, though they may have it for intervention purposes (e.g., autonomous vehicles).

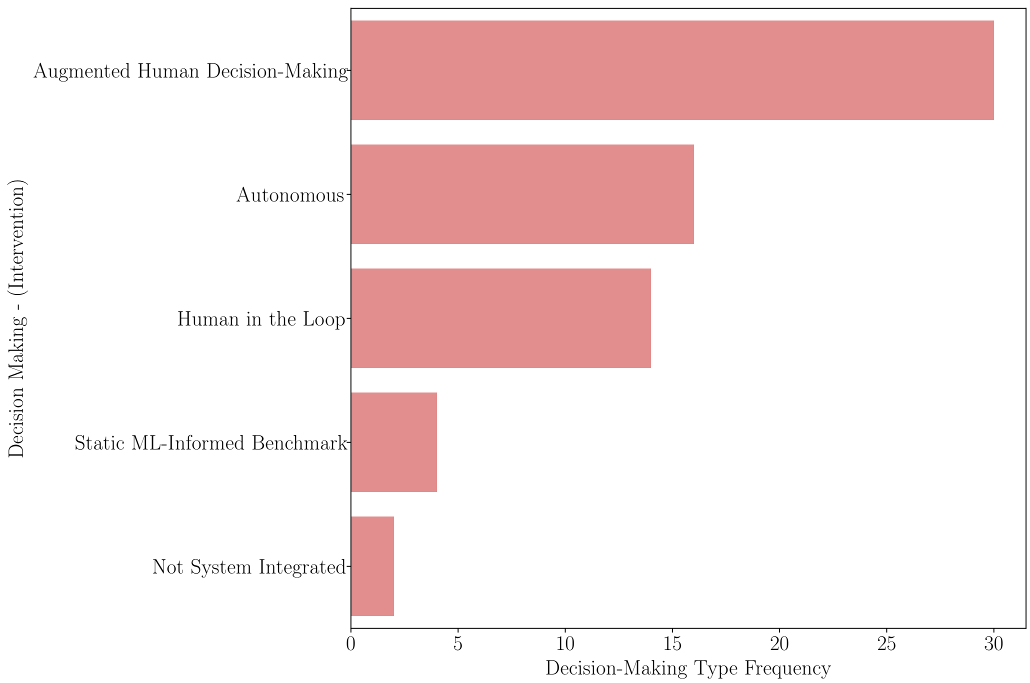


Figure 6: Decision-making types observed in final article subset.

Approximately 49% of the articles in our final subset considered explanation methods in some capacity. Of those that did, Figure 7 shows which explanation method they employed. In some instances, authors mentioned the importance of explainability, therefore considering it, but did not conduct any explainability analysis themselves. These articles were denoted a "Non-Specific" explainability type in Figure 7.

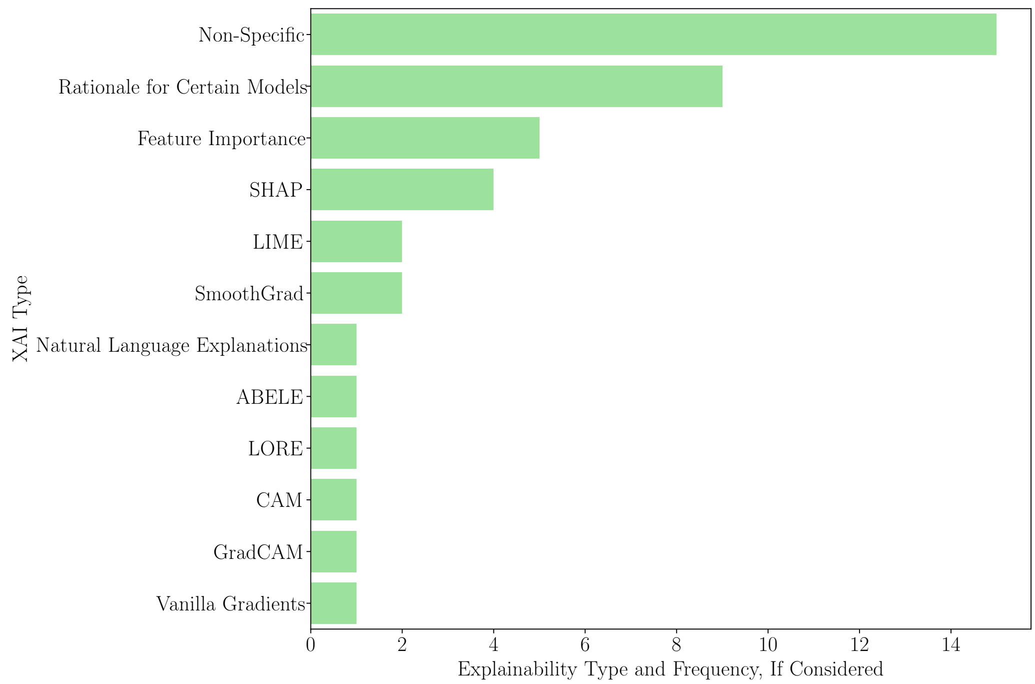


Figure 7: Explanation types observed in final article subset. “Non-Specific” indicates that explainability was considered, but no method was applied.

Figure 8 show the number of articles who have at least one contributing author from the countries shown. Of the final subset of articles, approximately 44% considered regulatory oversight. This distribution of countries is therefore intended to show how the regulatory oversight bodies might be distributed and to highlight the complexity of defining AI regulatory frameworks that span international boundaries.

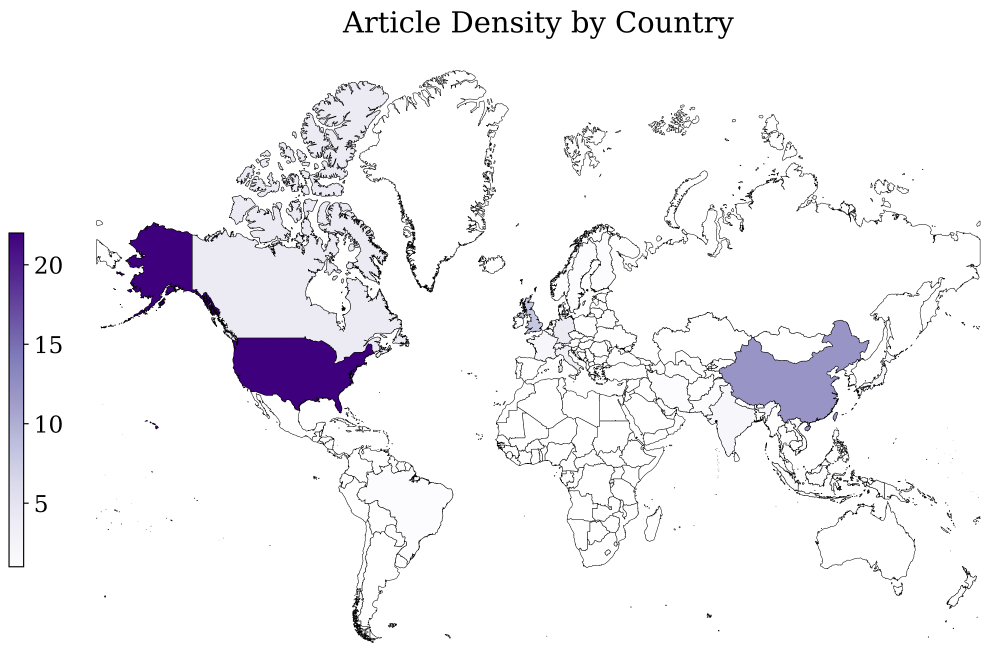


Figure 8: Number of articles in final subset who have at least one author from the listed country. "International Collaboration" article is excluded from this figure, see attached database for details.

Finally, Table 3 shows the distribution of articles by industry for the final subset.

Industry	Percent of Final Subset of Articles
Medical	59.72%
Autonomous Vehicles	18.06%
Nuclear	9.72%
Aviation	8.33%
Ethics	2.78%
Manufacturing	1.39%

Table 3: Final article subset distribution by industry.

4. Qualitative Results and Discussion

We will now discuss each of the major industries identified in our study, highlighting key takeaways from the individual industries and some papers that brought especially unique points to our attention. Then, we will examine the similarities and differences between the various industries to determine what AI features and findings were universally beneficial, and which industries have specific challenges and demands.

4.1. Medical

Medical papers comprised nearly 60% of our final subset of articles (see Table 3). As the largest industry in article numbers from both the query results and the final subset analysis, the medical industry has a lot to offer in terms of experience using AI systems and highly sensitive usage demands. AI in this industry generally has greater decision-making times and interfaces usually with highly skilled professionals (like doctors), but can also interact directly with patients. Due to its size and variety, we break down this group of papers into smaller categories.

4.1.1. Pharmacology

Eight out of the 43 papers categorized as medical considered applications related to the prescribing and dispensing of medication. The articles in this cohort emphasize that AI requires expert supervision, and the coordination of an expert and AI in pharmacological applications yields the best outcomes (mitigates safety risks). Though only two of these papers considered regulation, three considered model explainability. Interestingly, two of the papers that did consider explainability evaluated large language models (LLMs), which are widely considered the most opaque.

The aforementioned articles [10] and [11] both evaluate LLMs as medication safety-focused clinical decision support systems (CDSS), including prescription errors and drug-drug interactions. Both evaluated variants of Gemini, ChatGPT, and Claude. Ong et al. [10] evaluate the success rate of identifying “drug-related problems” (DRPs) in three scenarios: pharmacist alone, pharmacist with AI-assistance (co-pilot mode), and AI alone and found co-pilot mode to yield the highest success of DRP identification. Chase et al. [11] do not consider AI-pharmacist cooperation, and instead evaluate the LLM independently, still substantiating the finding that the LLMs perform suboptimally alone. They found the LLMs to make potentially harmful clinical recommendations up to 15.4% of the time (ChatGPT). Chase et al. [11] also raise concerns for the less overtly dangerous but still problematic scenario of suboptimal recommendations, such as flagging non-clinically relevant drug interactions, which can lead to alert fatigue over time. These reports highlight the importance of studying not only the pure accuracy of AI systems, but how their users interact with them.

In response to their findings, Chase et al. directly call for both external validation and benchmarking of medical AI against existing standards *and* model explainability techniques to address the black-box nature of LLMs before they are implemented into clinical practice [11].

However, not all "explainability" methods are created equal. Ong et al. [10] state that the LLMs tested (ChatGPT, Gemini, and Claude) offer something that many "explainable AI tools" cannot— "conversational interfaces [that] allow clinicians to ask questions and receive context-specific explanations in plain language." While the conversational nature of chatbots may seem like a more approachable "explanation," this logic can be dangerous as reasoning provided by a chatbot is *not* a true explanation and does not indicate *any* amount of model reliability, accuracy, or trustworthiness. We caution that this reasoning may lead to complacency and instead urge practitioners, regulators, and developers to consider genuine explainability and validation methods when applying these tools in practice. Explainability, in particular, can help the ultimate decision-maker (the pharmacist or prescriber) rationalize the CDSS recommendation, giving them grounds to dismiss or accept it with cause.

The remaining articles in the pharmacology subcategory use either classical ML techniques (decision trees and gradient boosting) [12], [13], [14], association rule learning [15], and CDSS built-in systems [16], [17]. Of these, Kieu et al. [12] considered decision trees explicitly for their inherent interpretability. From a deployment perspective, Liu et al. [17] interview pharmacists on trusting CDSS recommendations. They found that after consulting a CDSS, 78% of pharmacists did not adjust their dosing regimens due to "distrust and lack of understanding of the justification of the CDSS recommendations" among other concerns, despite a "generally positive attitude toward CDSS." This supports claims made by papers throughout this report that model trust is paramount to implementation and that this trust can be gained from explainability or "justification." Even if a co-piloted pharmacist-AI approach yields the safest medication prescription and dispensing, without pharmacist trust in the partnership, the accuracy improvements are moot.

4.1.2. Imaging and Diagnostics

Eight of the 43 papers evaluated in this section evaluated using AI for diagnostic and/or imaging techniques. Of these papers, specific AI techniques included convolutional neural networks (CNNs) for classification of glaucoma [18], LLMs for emergency medical advice [19], generative adversarial networks (GANs) and other methods for skin lesion classification [20], decision trees for cervical spine injury diagnosis [21], and CNNs for stroke onset time prediction [22]. Of the remaining three papers, two included the use of AI for prophylaxis efforts including preventing falls [23] and blood clots [24]. The last paper in this category discusses AI in radiology more broadly and offers methods to increase generalizability by overcoming overfitting and underspecification [25].

Of the articles considered in this subsection, three of them consider regulation and four of them consider explainability. Vieira et al. [18] model both, anticipating and acting on the needs of clinicians to understand model results and following regulatory transparency guidelines, such as the European General Data Protection Regulation (GDPR). They use retinography images on CNNs coupled with CAM, GradCAM, LIME, SHAP, vanilla gradients, SmoothGrad, and SCIM to evaluate the quality of the different explainability methods for a high risk clinical application (glaucoma). They found CAM-based methods most reliable to support diagnosis via clear identification of clinically relevant regions and intuitive visual interpretation.

While inherently interpretable models, like decision trees, can offer a clearer, more straightforward understanding of model logic, simpler models are not always an option, especially in the imaging and diagnostic realm. Vieira et al.'s ad-hoc explainability methods model acquiring expert trust in high-stakes applications [18], and it is likely that future work will emulate them in other highly sensitive domains as experts drive demand for rationalized AI recommendations.

4.1.3. Radiation and Other Treatment

Twelve ([16], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36]) of the 43 papers reviewed in this section evaluate the use of AI to design and carry out treatment plans with six specifically considering radiation treatment. The number of radiation treatment studies that employ AI alone represents the perceived utility of AI at improving highly sensitive treatment outcomes. Of the twelve papers considered in this section, seven of them considered some aspect of explainability, but only two of them considered regulation, for insulin dosing and [16] maxillofacial trauma surgery [36]. Of the radiation treatment papers considered, nearly all employed classical ML methods, including decision trees, random forests, support vector machines, and gradient boosting. Only one paper [28] considered UNets, a type of CNN, exclusively, and another paper considered neural networks in addition to classical methods [32]. Six [29], [30], [32], [33], [34], [31] of these seven papers either conducted feature importance studies on these classical methods or rationalized using an inherently interpretable model for explainability purposes, highlighting the appeal of simple-as-possible models.

As an outlier, Komorowski et al. [26] used reinforcement learning as the primary learning method for intensive sepsis care, but still used random forests to determine clinically interpretable reasoning (importance of model parameters). The majority of these studies indicate a preference towards simpler, more comprehensible models, and an even greater majority employ explainability for highly sensitive treatment methods.

Most of these papers considered AI to find some threshold radiation dose that is sufficient for treatment success while minimizing harm and secondary radiation side effects to the patient. One of the papers that did not consider explainability outright, did perform an uncertainty analysis on their UNet model for dose prediction to treat prostate cancer, which still emphasizes a need for some method to establish model trust [28]. Overall, the trend in treatment papers towards explainable AI-decision-support methods supports what we saw in pharmacology– that experts need rationalized recommendations before implementation. Most of the papers in this section focused on doctors administering treatment to patients, therefore overseeing the AI models in question. The one paper [16] that interfaced directly with patients was validated through clinical studies and regulated by the FDA, showing the substitute role (in this case, empirical) validation can play in establishing model trust.

4.1.4. Legal and Regulatory

Eight of the 43 articles considered under the "medical" umbrella directly concerned legal and regulatory factors. Unlike other subsets, these papers generally do not evaluate a specific type of AI. Instead, they describe the ethics, liability, and regulatory considerations that govern these technologies in sensitive domains. Of the eight, five legal/regulatory papers consider explainability in their discussions. In terms of governing bodies concerned, three papers are from E.U. member states, two are from the U.S., two are from the United Kingdom, and one is from Canada.

Of the articles we considered in this subsection, we identified common themes of trust, trustworthiness, and liability. The former are distinguished by Jones et al. [37] following Onora O'Neil's previously established trust framework, trust is something you give and trustworthiness is how deserving of your trust an entity is. O'Neil emphasizes that trust is only dangerous when placed in an untrustworthy entity. Further, she notes that *"only beings with agency are truly capable of being trusted"* [37].

We can define the individuals issuing trust and those assessing trustworthiness as separate cases. In the industries considered in this review, when the public is a direct user of an AI-based decision-making system, we must assume they do not have the expertise to assess trustworthiness of that system. Therefore, these individuals must trust the judgment of some other trustworthiness assessor. For safety-critical industries, as an example, this assessor could be the regulator. In other words, the public places their trust in the regulator, not in the AI itself.

Alternatively, if the direct user of an AI-based decision-making system *does* have the expertise to assess that system's trustworthiness (such as a medical professional), then they can decide on a case-by-case basis when they are willing to accept its recommendations, and they can serve as a barrier to the less knowledgeable public. In this case, the public places their trust in the experienced professional, not in the AI itself.

These two cases also extend into AI use in treatment and in diagnosis, where the former may be used by an expert or non-expert, and the latter only with experts. The responsibility of the regulator differs in these cases, where in treatment they may authorize that a system may be trusted outright, but in the diagnosis they create frameworks to support the decision-maker's assessment of model trustworthiness. Counterintuitively, over-regulating diagnostic AI could increase patient risk by taking liability away from the professional using it, resulting in complacency. The regulatory authority decision of federal agencies in *Loper Bright*, which we discuss later in this section, further complicates this dynamic.

In reality, the cases of trust and trustworthiness outlined above may not be so clear-cut. Trust may fall reasonably back on model developers, users, procurers, etc. Following O'Neil's logic, if a system possesses no agency, it cannot be trusted. The liability question then becomes: *Who is responsible for the consequence of this system?* This is the debate of many legal experts and should be considered by all participants in an AI system, especially as laws and regulations surrounding these technologies change. As a potential solution, Lagiola et al. [38] and Labkoff et al. [39] suggest that unique regulations should correspond to classes of AI automation levels to define the shared level of responsibility and liability between the AI system and the individuals involved in its use.

The current licensing system in the U.S. is more vague than that and leaves many of the aforementioned questions up to the courts. Instances where AI-assisted decision systems are used by a skilled professional are less regulated and instead rely on their expert judgment (and malpractice insurance). Treatment-based medical AI systems, on the other hand, mainly qualify as medical devices, and can be regulated as such in the U.S.

Briefly, the FDA offers three main pathways for licensing depending on the level of rigor of their review. This includes Pre-Market Approval (PMA), *De Novo*, and 510(k), in increasing order of level of rigor. *De Novo*, the newest, is a hybrid pathway that offers neither all the litigation protections of PMA nor the speed of 510(k). Though outcomes of this pathway remain legally uncertain, AI/ML-enabled device manufacturers use the *De Novo* pathway about three times more often [40], suggesting these companies prefer to opt for uncertainty in future liability defense in exchange for expedited approval. This may concern the public as they rely on the FDA's trustworthiness assessment for these (and non-AI) devices. A more thorough approach may include rigorous validation techniques and/or a unique licensing pathway for a unique set of technologies, especially for treatment devices that necessarily elicit trust.

Other legal complications of medical AI may be less obvious. McCradden et al. [41], Kiseleva et al. [42], DeMicco et al. [43], Habli et al. [44] and many others emphasize the ethical concerns of bias in AI systems. However, biased AI systems may play into the aforementioned regulatory conditions as well. Gerke et al. [40] describe that when licensing a medical device, it is licensed as-is. However, "continuous learning" or even discretely retraining an AI system may worsen efficacy for certain patient groups [40]. Shah et al. [45] extend this concern to broader changes post-market release. Failure to warn users of these changes could leave manufacturers liable, even if their original model was protected. Changing efficacy across patients groups, however, may be difficult to assess without formal validation techniques. Thus, aside from being ethically valuable to consistently validate and evaluate models for bias, we also find legal motive to embed this into system design.

When validation methods are not perfectly accurate, explainability methods offer a second layer of protection to the patient, doctor, and AI manufacturer. As explained by DiMicco et al. [43], "AI systems should be developed to enable humans to understand their actions and the underlying logic in order to increase transparency and minimise the risk of bias or error." Some jurisdictions have already begun to require aspects of explainability, such as in the European Union's General Data Protection Regulation (GDPR) Articles 13 and 14, which require "meaningful information about the logic involved, as well as the significance and the envisaged consequences of such processing" when decision-making is automated [38]. Not all jurisdictions offer such a proactive regulatory approach.

Importantly, safety critical domains in the U.S. should not rely on regulations to protect them from potential faults of AI-based decision support systems, because the authority of regulatory agencies in these areas is still legally uncertain. As highlighted by [40], the 2024 Supreme Court case *Loper Bright Enterprises v. Raimondo* and the overturning of the *Chevron* doctrine affects how regulatory agencies can claim authority over these decision support tools. Prior to *Loper Bright*, courts were expected to defer ambiguous jurisdictional authority to the interpretation of the regulator; now these ambiguities must be interpreted independently by the courts. This complicates the ability of American regulatory bodies to regulate AI-tools, such as CDSS. This may extend to other industries in which experts might use AI to make "diagnoses," such as in nuclear power or other industrial operations. Practically, this shifts responsibility toward professionals using AI-based decision support, and encourages independent validation and interpretability efforts instead of relying on federal oversight.

Having established how regulation relates trust and trustworthiness, we can now consider how developers and users each evaluate these AI systems– and the consequences of their misalignment. We can accomplish this first by understanding O'Neil's three constituents of trust, as described by Jones et al. [37]: trust in truth claims, trust in commitment, and trust in competence. The first is an empirical evaluation, *Is this entity likely to give me a factually correct answer?*, whereas the second two are normative, *Is this entity likely to follow through on their promises?*, and *Is this entity likely to execute the promised action successfully?*.

Developers typically focus on the empirical arm of these, as evidenced by the quantitative way we evaluate AI systems: *given 1,000 scenarios, how often will my system predict the correct answer?* Yet, we seldom consider the other two thirds of how people perceive trust when we set out to develop a "trustworthy system." This misalignment can set users up to overtrust a system that has obvious limitations to the developer.

The tendency for individuals to place undue trust in these systems by focusing on the normative elements is evident. After considering the perspective of clinicians incorporating AI into their practice and the empirical trust issues they experience, Jones et al. [37] also consider how patients feel about their doctors using AI. They summarize several studies that asked hypothetical patients how they would view various outcomes when their doctors did or did not use AI systems. Overwhelmingly, they found that patients at worst had no difference of opinion when AI is used, and at best, view their doctors' decisions more favorably, even if the outcome is still unfavorable, showing that AI may serve as a "protective shield" for clinicians [37].

There are, of course, some caveats to this: the studies were hypothetical and generally assumed that the AI in question had a known superior diagnostic accuracy than the doctor. Jones et al. conclude that patient favorability towards using AI stems from higher normative trust in two ways: one, that AI is both reliable (machines tend to be more predictable than humans) and two, that AI is competent (on the basis of mass amounts of training data) [37]. Both the readiness of patients to (blindly) trust AI and the clinician's concerns about its empirical accuracy further support the case for co-piloted AI-clinician systems and raise red flags for autonomous AI healthcare providers.

Despite public overconfidence in AI devices, domain experts tend in the opposite direction. In a trial run of AI-assistance at the U.S. FDA to analyze causality in individual case safety reports (ICSRs), reviewers rejected the AI's analysis in most cases because "...the approach did not offer sufficient explanation or flexibility to accommodate information external to the ICSR (e.g., reviewer's knowledge)" [46]. Further, "[b]ecause of the lack of trust in the software tools, any efficiency gains... were not perceived as making it easier for the safety reviewers..." [46]. This highlights a case where domain experts not only question the AI's truth claims, but also its competence given a lack of sufficient data. The critical feature of this scenario is *the expertise of the reviewer*. A less knowledgeable individual may more readily accept the claims of an untrustworthy system, simply because they did not know better. This underscores the importance of "human-in-the-loop" systems and the urgent need to maintain the proficiency of subject-matter experts.

Although experts remain critical of unexplainable AI, developers and regulators should still consider ways to counter "automation bias," the tendency of users to err on the side of the computer [37]. In other words, how can we prevent domain experts from doubting their own judgment when AI suggests a (potentially wrong) alternative? *How can we encourage them to "trust their gut?"* Holding professionals accountable for their decisions will likely encourage this. Still, how can we ensure that experts are protected when they rely on their own professional expertise? Explainability measures may mitigate automation bias by allowing practitioners to critically analyze the reasoning of the AI, creating a situation more similar to a collegial disagreement. Future work may consider implementing guardrails in software design for these known shortcomings of AI-assisted decision-making systems, and regulation should support trustworthiness assessments while maintaining the liability of the professional.

4.1.5. Qualitative Analyses and Roadmaps

The final medical category consisted of six qualitative papers that underscore our finding that expert trust is integral to the success of an AI-supported decision-making system. Two papers focused directly on healthcare professionals' perspectives [47], [48] while another focused directly on patient perspectives [49]. The remaining three make broad recommendations on what AI in healthcare ought to look like, all of which included addressing risk of bias, validation, regulation, and explainability as key issues [39], [50], [51].

The papers in this section uniquely highlight the importance of cross institutional collaboration. Labkoff et al. [39] propose an CDSS-incident reporting platform to mitigate the repetition of mistakes and standardizing documentation to inform clinicians of tool limitations [39]. Similarly, Taylor et al. [51] propose mitigating bias through model "validation across institutional datasets." Unfortunately, sensitive applications like healthcare also come with sensitive data. This is a challenge we can expect across sensitive domains that may be addressed with organizational efforts or federal guidance. Future work may identify safe, anonymized methods to do so.

Finally, the interview-based perspective papers offered insight from both sides of medical practice. They found medical professionals, including doctors, nurses, staff scientists, and administrators all similarly concerned about model recommendations without rationale and the risk of bias, especially because they felt ultimately liable [47], [48]. Alternatively, the patients interviewed in Fransen et al. [49], showed a clear preference for using AI versus not to aid radiologist's judgment of their prostate cancer imaging, but most

(85%) preferred radiologist oversight [49]. Fransen et al. [49] acknowledge the limitations in their study by referencing a different outcome found by an older study on women's acceptance of AI in mammogram interpretation that found a notably lower level of acceptance. This difference could be attributed to a variety of factors other than gender, including the year of publication (2025 for Fransen et al. and 2021 for the mammogram study), but it highlights the importance of not just evaluating bias within the model but also understanding how different individuals may view these systems.

4.2. Autonomous Vehicles

Autonomous vehicles comprised just over 18% of our final subset of articles, our second-largest category. These papers ranged from focusing directly on ethics [52], [53], [54], [55], to validation [56], [57], explanations [58], [59], legal implications [60], [61], and pure accident avoidance [62], [63], [64]. Broadly looking at the subcategories alone highlights some unique distinctions from the other industries in this review. First, the overwhelming number of ethics-focused papers. Unlike in medical contexts, autonomous vehicles often face two-body ethical dilemmas– *who should be spared in an unavoidable collision scenario?* Further, direct calls for validation and regulation are prolific.

Autonomous vehicles, as their name implies, do not enjoy the protections of "human-in-the-loop" systems, which is what most medical AI operates under. Nor do the users of these AI-driven systems reap the benefit of a certified expert reviewing each and every decision. Instead of the AI offering a suggestion and an experienced professional using their expertise to decide whether to take it, autonomous vehicles make decisions nearly continuously and rely on human intervention *only* if something goes awry. This removes a barrier of protection against unreliable systems, making liability issues more complicated, especially when defining "human intervention." How could someone intervene a system designed to relieve their attention span? Tian et al. [61] approach this argument by promoting proactive insurance involvement and liability rules that encourage manufacturers to invest in safety. Tian et al. [61] explain this in the context of the Hand Formula– if the burden (of safety innovation) is less costly than the probability (of an accident) times the loss (of that accident), it is negligent for the manufacturer *not* to take that safety precaution.

To make the safety designation even more challenging, although many of the AI systems evaluated in this paper are "narrow AI," meaning they are trained for a specific task (the opposite of general AI), autonomous vehicles face a wide variety of scenarios, and can struggle to learn edge cases that have severe consequences (as seen in [56]). This, combined with a lack of general oversight, points to a major need for validation methods that can serve as the basis of verifying the safety of these systems to the public. This is perhaps why we see the most validation studies in this subcategory– two [56], [57], versus one for aviation [65], and none for any other category.

Generally, we denote validation and explanation as distinct but similarly useful practices for AI-supported decision-making. In the context of autonomous vehicles, however, the latter is less effective. While explanations can be helpful in understanding the rationale of an autonomous vehicle's decision-making, they are less useful than CDSS explanations because the window of time between an autonomous vehicle's data acquisition, modeling calculation, and decision is much shorter. In other words, one would probably hope that a doctor takes more time to decide a patient's treatment plan than they take to change lanes on their way to work. The same time restriction on those decision-making scenarios applies for autonomous vehicles, so model explanations are not particularly useful for human intervention.

Counterintuitively then, two papers in this category focus *primarily* on developing model explanations [58], [59]. In these reports, explanations play other roles than rationalizing real-time decision-making, like evaluating model performance before deployment [58] or understanding trust dynamics with human operators [59]. We do caution that like Ong et al. [10], Feng et al. [58] generate somewhat "pseudo-explanations" by jointly predicting decision-making actions and natural language explanations using natural language processing. These are not true explanations that glean insights on the inner-workings of a model, so they cannot be given the same authority as feature importance from a random forest, for example. The attempts made towards autonomous vehicle explanations indicate a diversion from studies with greater decision-making windows and experts at the helm of the final decision calls, but we generally find explanation methods less useful in the context of autonomous vehicles than in other domains.

4.3. Nuclear

Nuclear systems comprised nearly 10% of our final subset of articles. This subset varied in terms of nuclear application, with most studies we considered toeing the line with predictive maintenance, a classification we excluded from this analysis. All papers related to the operation of nuclear power plants in this section, six of the seven papers, used traditional machine learning methods (i.e., decision trees, random forests, feedforward neural networks) plus more complex models, like Bayesian belief networks and long-short term memory networks. The only paper unrelated to nuclear power plants focused on nuclear crisis management and used generative AI in its analysis [66]. Holmes et al. [66] stood out as an outlier in this category, not only because the application area broadly differed, but because it focused on using generative AI to avoid the pitfalls of human emotion in high-stakes political decision-making scenarios. In most of the studies we consider in this work, AI is used to support decision-making from a logical, data-based perspective, but in this case, the AI is used to assist in a somewhat emotional capacity. While this paper met all of our inclusion criteria, it serves a small audience and is basically unregulated. Still, it emphasizes a need for humans to remain in the loop for expert-level decision-making.

The nuclear-power-focused papers in this section considered using AI for reactor pressure vessel design [67], severe accident scenario propagation [68], cooling water intake reliability [69], fault detection of sensors [70], accident diagnosis [71], and human error detection [72]. Of these, three considered explainability methods, one inherent to the AI model used (decision trees) [68], one using a post-hoc method known as SHAP [72], and one just as a general concept that *should* be incorporated [69]. None of the methods in this category were deployed, and only one considered regulation [70]. Since nuclear power is a highly-regulated, highly-skilled industry, we find the limited attempts at developing explanations and considering regulatory compliance indicative of the state of the field. The systems considered in this category were all subject to human oversight; no papers suggested a fully autonomous system. This means that all AI outputs will need a human go-ahead before making decisions. Without adequate rationale, we can expect nuclear power plant operators and engineers to be just as unwilling to blindly trust black-box recommendations as the medical professionals in Section 4.1. Future work on developing explanatory methods, validation techniques, and regulatory compliance strategies will therefore be critical in the nuclear power space.

Further, unlike other industries in this review, nuclear power faces the unique challenge of extremely limited data availability. This is evidenced by the fact that three of the six nuclear power papers use synthetic data, while the other three rely on mechanical sensors or static measurements. The use of synthetic data adds another layer of epistemic uncertainty to the model results that is avoidable in more data-prolific applications. The use of synthetic datasets will likely require additional validation to meet regulatory standards for safety-critical decision-making in nuclear systems. This barrier is characteristic of the field— almost all the papers in this section mention limitations from lack of data. As such, we also observe most of these studies are "preliminary" or "proof-of-concept" where other industries have already deployed and sometimes validated their models.

Finally, the studies in this subsection are geographically concentrated to the U.S., South Korea, China, and Sweden, likely as an effect of the more limited global distribution of nuclear power compared to other technologies, like medicine and aviation. The size and distribution of this industry, combined with its limited data and high risk nature makes AI research more difficult to transition to a deployed technology. Future work in this area might focus on the regulatory challenges of implementing AI, as well as a way to address data limitation issues.

4.4. Aviation

Aviation comprised 8.33% of our final subset of articles (see Table 3). Of the aviation papers, three were specifically aimed at predicting accidents and incidents, two during flight [73], [74], and one during scheduling [75]. The three remaining studies looked at automated airport security detection [76], a Bayesian validation method for black-box systems that is currently employed by the FAA [65], and a roadmap for AI in flight advisory systems [77]. The airport security screening and the validation study are both deployed systems, while the rest of the studies present pure research. Of the six papers, only one, Cankaya et al., considered explainability [75].

Aviation AI targets low-probability, high-risk events that are generally supervised by experienced professionals (e.g., pilots). Due to the varying ways AI can be used in aviation, some scenarios face the same time-restriction challenge as autonomous vehicles, but not all. That said, for applications with human agency and sufficient time for review, explainability methods may be useful. Cankaya et al. [75] use this logic to justify needing interpretable models for the managerial decision-making in aviation– creating a zero-accident ecosystem requires knowing the root causes of *potential* accidents before they happen. While "black-box" systems may provide insight on the frequency of failure, they cannot provide likely causes to enable prevention. Interpretable models may therefore fill this gap and allow rationalized decision-making for time eligible scenarios.

Other studies in this group [77], [74], [78], and [76] use AI to improve safety in various aspects of aviation (via flight advisory systems, hard landing warnings, engine flameout predictions, and accurate security screenings, respectively), but they do not evaluate model explainability or perform validation studies on their models. Such levels of model opacity and performance uncertainty leave their frameworks far from implementation from a regulatory perspective, except for [76], which evaluates an already deployed system that is separate from the mechanical functionality of the aircraft and therefore subject to different scrutiny.

Notwithstanding their opacity, many non-interpretable models (such as those listed above) perform extremely well when tasked to predict an upcoming incident. While such models are obviously valuable, understanding their reliability, if not their exact inner-workings, is critical for deployment. Moss et al. develop a way to accomplish this [65]. They model the failure probability estimation for autonomous flight systems using a Bayesian framework that "...iteratively fits a probabilistic surrogate model to efficiently predict failures" [65]. As the authors state, this system is actively involved in an FAA certification process. As such, other industries might apply a similar protocol to validate AI systems to their respective regulatory bodies. Over all industries, aviation contained one of the few validation studies (though other industries call for validation, such as in [41], which calls for "model auditing"). Though nearly all the industries considered in this review are regulated to some extent, the lack of validation studies, something we would typically expect for implementing new technologies into highly regulated fields, are evidently lacking.

It is not surprising, however, that the most rigorous validation study we found is from aviation, as the aviation industry is widely known for its safety-first culture. In fact, during our analysis, we found other industries explicitly reference aviation's success in implementing such a strong safety culture and their goal to replicate it [50]. Fuchs et al. [50] emphasize the importance of validation studies like [65] when they call for a graded approach to regulating medical-AI as part of a larger roadmap to mimic aviation's historic tactic to best safety practices.

4.5. Other Industries

The three remaining papers in our review included two legal/ethics papers [79], [80], and one paper on manufacturing defects [81]. While the latter is a more traditional machine learning paper, it emphasizes explainability when predicting void formations during welding, a phenomenon that can lead to unreliable structures [81]. Of the former, Maskur et al. [80] mainly discusses global variations in current AI regulations, stating that the E.U. has taken the most direct, proactive approach, while the U.S. waits for precedents to arise through the courts. We see these differences in the medical space, as described in Section 4.1.4, and we suggest that future work consider how to remediate these differences when technologies cross national borders.

After all the trust concerns described in this review, it seems natural to question if AI is even worth implementing to our most sensitive domains. Is the status quo good enough? Safer, even? Cerf et al. [79] present a response to this very concern and cautions readers that while fears surrounding AI are well-founded, humans have historically proven problematic as well. While Cerf's claims likely aim to address existential concerns of generative AI, we can still heed their warning that we not let risks surrounding AI "...distract from human beings' current proficiency at carrying out the threats attributed to technology" [79]. As such, we can take away from this is that rather than run from a powerful technology, we ought to design, implement, and regulate it with a safety-first approach. By completing this review, we have identified general rules to accomplish this goal, which we describe the following section.

5. Conclusion

From this review, we find two key principles that remain universally true across industries. The first is related to explainability. We find that explainability techniques only matter if the user has both the *time* to consider the rationalization and the *expertise* to review it. In other words, if the interfacing user is not a domain expert, developers and regulators should not assume that the user has the necessary background knowledge or understanding to adequately assess the quality of the AI's decision given an explanation. Likewise, even a domain expert cannot possibly evaluate an explanation instantaneously. In scenarios with limited user expertise and/or limited evaluation time, it is necessary to have robust regulation underpinned by model validation techniques to protect users. Validation is always a good idea but is strictly necessary when AI tools interface with inexperienced or time-pressed individuals. Model explanations are both unhelpful and insufficient in these contexts. Some examples of this may include wearable medical devices, like glucose or heart rate monitors that encourage decision-making without physician review, or broadly, autonomous vehicles.

Second, in scenarios where the user has both sufficient time and expertise to review a model explanation, the explanation is incredibly useful and practically necessary for acceptance. Validation in these scenarios is, of course, reassuring. However, when the decision is ultimately made by a highly skilled professional, they bear the responsibility and should also be liable for the decision they made. Overly prescriptive regulation for decision-support systems used by highly skilled professionals encourages complacency. Such complacency would be detrimental to quality of care in domains where we rely on professional oversight to reap the benefits of co-piloted AI-human systems. The studies in this report broadly indicate that AI-assisted decision-making yields superior results when paired with a critical expert. Absolving these experts of their authority would certainly worsen outcomes. In that vein, we should also protect the opinions of professionals when they disagree with AI-recommendations. If patients can reliably win lawsuits against doctors for diverging from CDSS recommendations, then we have effectively nullified their professional opinions and thereby relinquished oversight of CDSS systems. Regulators, lawmakers, developers, and users should all be aware of this risk.

5.1. Study Limitations

Nearly all studies in this report found that AI technologies "improved outcomes" in some way. This is expected due to publication bias– null results simply do not get published with the same frequency as successful protocols. Second to this, of the 72 studies considered, only 21 involved what we considered a "deployed system," meaning it was tested on real, critical scenarios. The remaining papers were either not deployed (38) or non-applicable as they were qualitative or roadmap papers (13). This means that even a broad systematic review like this cannot conclusively say if AI improves outcomes in sensitive industries. Future work might consider exclusively looking at deployed AI in a specific domain for more general conclusions on lives spared by AI, but even this effort will suffer the effects of publication bias. We also observe a chicken-and-egg problem in this facet—sensitive industries will be hesitant to deploy AI systems for safety-critical applications until they are proven safe and/or better than the status quo, yet the only way to evaluate this for a complete black-box technology is to study the outcomes of such a system on real scenarios. The solution is called for in papers throughout this review and explicitly demonstrated by Moss et al., a model-agnostic validation technique for black-box AI that can be reviewed by regulators [65].

That said, insufficient regulation may render the most effective validation techniques futile. This matters most for non-expert facing technologies where robust regulation must establish the validation criteria for model developers to apply.

Next, the authors of this review do not have expertise in all the fields evaluated. For this reason, there may be nuances in the subdomains that we missed. However, we believe the importance of cross-industry learning supersedes this potential limitation.

This review strictly sourced material from published academic articles on Web of Science and Scopus. As such, our review of relevant regulations for the various industries covered is necessarily incomplete. This, in and of itself, may underscore the disconnect between software developers, industry practitioners, and regulators. We urge each party to make efforts to close this gap to promote the safe use of this technology. Papers such as this serve as a first-step in that direction.

Finally, the search terms we employed in our methodology may have missed some critical papers. To manage our review capacity, we were forced to create and limit our terms to the most-likely relevant papers, though we recognize that some have fallen through the cracks. Because of this, the conclusions drawn in this report should represent general trends in the literature, but it is likely that counterexamples exist, especially as the state of AI research rapidly progresses.

5.2. Future Work

Though we describe opportunities for future work throughout this report, a few stand out. First, we observed data limitation issues across industries in this review. Future work should seek to improve the quality, accessibility, security, and cross-organizational sharing of data for sensitive applications. More broadly, future work should evaluate the security risks of AI-based decision-making in sensitive industries to both inform regulation and help guide data limitation efforts. From a policy perspective, future work should evaluate how international technology can manage varying national regulations in a way that emphasizes user safety. Likewise, future work should extend on the findings of this report to evaluate the nuances of AI safety in industry-specific applications. Domain-specific regulatory bodies may require this detail for effective policies. Finally, future work must continue to explore validation and explanation techniques. The quality of these techniques will underpin trust in this powerful technology.

6. Acknowledgment

The authors would like to thank Riley Fisher for his feedback and suggestions during review, Leo Tunkle for his help with early abstract screenings, and Dr. Sam Dotson for his feedback during methodology development.

This research is funded by the U.S. Nuclear Regulatory Commission's University Nuclear Leadership Program for Research and Development, award number 31310024M0013. The first author (N. Panczyk) received sponsorship through the National Science Foundation's Graduate Research Fellowship Program.

7. CRediT Author Statement

- **Nataly R. Panczyk**: Conceptualization, Methodology, Software, Validation, Formal Analysis, Visualization, Investigation, Data Curation, Writing - Original Draft.
- **Majdi I. Radaideh**: Conceptualization, Methodology, Data Curation, Formal Analysis, Resources, Funding Acquisition, Supervision, Project Administration, Writing - Review and Edit.

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